Module 6 — Particles and Medical Physics

This module covers capacitors, electric fields, electromagnetism, nuclear and particle physics, and medical imaging.

6.1 Capacitors

Capacitance (C): The charge stored per unit potential difference across a capacitor. A capacitor stores energy in an electric field between its plates.

$$C = Q / V$$

Units: farads (F), where 1 F = 1 C V^{-1}. Typical values: μF (10^{-6}), nF (10^{-9}), pF (10^{-12}).

Energy Stored

$$E = ½QV = ½CV^2 = ½Q^2/C$$

This energy is stored in the electric field between the plates. On a Q-V graph, the energy stored is the area under the graph (a triangle for a capacitor, since Q ∝ V).

Capacitors in Series and Parallel

	Series	Parallel
Charge	Same on each: $Q_{total} = Q_1 = Q_2$	Adds up: $Q_{total} = Q_1 + Q_2$
PD	Adds up: $V_{total} = V_1 + V_2$	Same across each: $V_{total} = V_1 = V_2$

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Capacitance	$1/C_{total} = 1/C_1 + 1/C_2$	$C_{total} = C_1 + C_2$

Exam Tip: Note that capacitors in series combine like resistors in parallel, and vice versa. Capacitors in parallel have a larger total capacitance; in series the total is smaller than the smallest individual capacitor.

Charging and Discharging

Discharging through a resistor:

$$Q = Q_0 e^{-t/RC}$$

$$V = V_0 e^{-t/RC}$$

$$I = I_0 e^{-t/RC}$$

All three quantities (charge, voltage, current) decay exponentially during discharge.

Charging through a resistor:

$$Q = Q_0(1 - e^{-t/RC})$$

$$V = V_0(1 - e^{-t/RC})$$

Charge and voltage grow towards their maximum values. Current still decays: $I = I_0 e^{-t/RC}$.

Time Constant

Time constant (τ): τ = RC. The time taken for the charge (or voltage or current) to fall to 1/e (≈ 37%) of its initial value during discharge, or to rise to (1 − 1/e) ≈ 63% of its final value during charging.

- After 1τ: value falls to 37% (or rises to 63%)

- After 5τ: capacitor is considered fully charged/discharged (< 1% remaining)
- To find τ experimentally, plot ln(V) or ln(Q) against t. The gradient = −1/RC.

6.2 Electric Fields

Electric field: A region of space in which a charged particle experiences a force. Field lines go from positive to negative.

Coulomb's Law

$$F = kQ_1Q_2 / r^2 = Q_1Q_2 / (4\pi\varepsilon_0 r^2)$$

Where $k = 1/(4\pi\varepsilon_0) = 8.99 \times 10^9$ N m^2 C^{-2}, ε_0 = permittivity of free space = 8.85×10^{-12} F m^{-1}. Like charges repel; unlike charges attract.

Electric Field Strength

Electric field strength (E): The force per unit positive charge at a point in the field.

$$E = F / Q$$

Radial field (around a point charge or charged sphere):

$$E = kQ / r^2 = Q / (4\pi\varepsilon_0 r^2)$$

Uniform field (between parallel plates):

$$E = V / d$$

Where V = potential difference between the plates (V), d = separation (m). Units: V m^{-1} or N C^{-1}.

Electric Potential

$$V = kQ / r = Q / (4\pi\varepsilon_0 r)$$

Units: V (J C^{-1}). Unlike gravitational potential, electric potential can be positive or negative depending on the sign of the charge.

Electric potential energy:

$$E_p = kQ_1Q_2 / r$$

Exam Tip: Electric and gravitational fields have many analogies. However, gravity is always attractive while electric forces can be attractive or repulsive. Also, gravitational potential is always negative, while electric potential can be positive or negative.

6.3 Electromagnetism

Force on a Current-Carrying Conductor

$$F = BIL \sin\theta$$

Where B = magnetic flux density (T), I = current (A), L = length of conductor in the field (m), θ = angle between the conductor and the field. Use Fleming's left-hand rule: First finger = Field (B), seCond finger = Current (I), thuMb = Motion (F).

Force on a Moving Charge

$$F = BQv \sin\theta$$

Where Q = charge (C), v = velocity (m s^{-1}), θ = angle between velocity and field. The force is always perpendicular to both v and B, so a charged particle moving perpendicular to a uniform B field follows a circular path (magnetic force provides centripetal force).

$$BQv = mv^2 / r \;\Rightarrow\; r = mv / BQ$$

Magnetic Flux and Flux Linkage

Magnetic flux (Φ): The product of the magnetic flux density and the area perpendicular to the field.

$$\Phi = BA\cos\theta$$

Where Φ = magnetic flux (Wb), B = magnetic flux density (T), A = area (m^2), θ = angle between B and the normal to the area.

Flux linkage = NΦ (for a coil of N turns). Units: Wb (or Wb turns).

Faraday's Law

Faraday's law: The magnitude of the induced EMF is equal to the rate of change of magnetic flux linkage.

$$\varepsilon = -N\,d\Phi/dt = -d(N\Phi)/dt$$

Lenz's Law

Lenz's law: The direction of the induced EMF (and hence the induced current) is such that it opposes the change in flux that produced it. This is a consequence of conservation of energy.

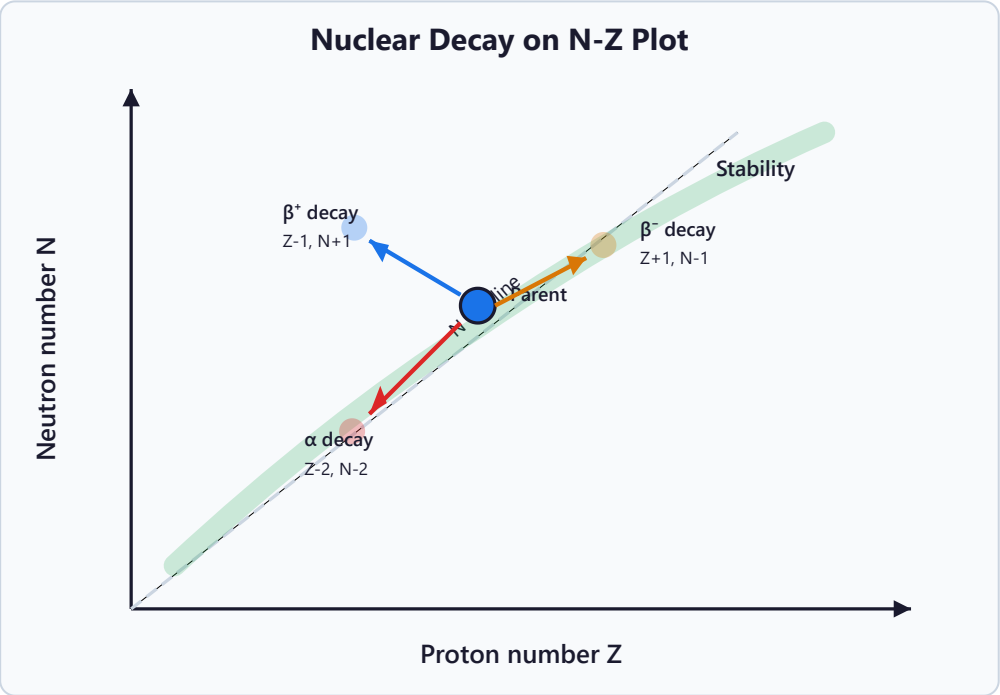
The negative sign in Faraday's law represents Lenz's law.

Exam Tip: To determine the direction of induced current, ask: what change in flux is occurring? The induced current must create a magnetic field that opposes that change. If flux is increasing, the induced field opposes the external field; if decreasing, it supports it.

6.4 Nuclear Physics

Radioactive Decay

Property	Alpha (α)	Beta-minus (β^-)	Gamma (γ)
Nature	Helium nucleus (^4_2He)	High-speed electron ($^0_{-1}e$)	High-energy EM radiation (photon)
Charge	+2e	−e	0
Mass	4 u	~1/1836 u	0
Penetrating power	Low (stopped by paper / few cm of air)	Medium (stopped by ~3 mm aluminium)	High (reduced by several cm of lead)
Ionising ability	Strongly ionising	Moderately ionising	Weakly ionising
Deflection by E/B fields	Deflected (positive charge, large mass)	Deflected opposite to α (negative, small mass — more deflected)	Not deflected



Decay Equations

Alpha decay:

$$^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}\alpha$$

Beta-minus decay (neutron → proton + electron + antineutrino):

$$^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}\beta + \bar{\nu}_e$$

Beta-plus decay (proton → neutron + positron + neutrino):

$$^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + {}^{0}_{+1}\beta + \nu_e$$

In all nuclear equations, mass number (A) and charge number (Z) are conserved.

Radioactive Decay Laws

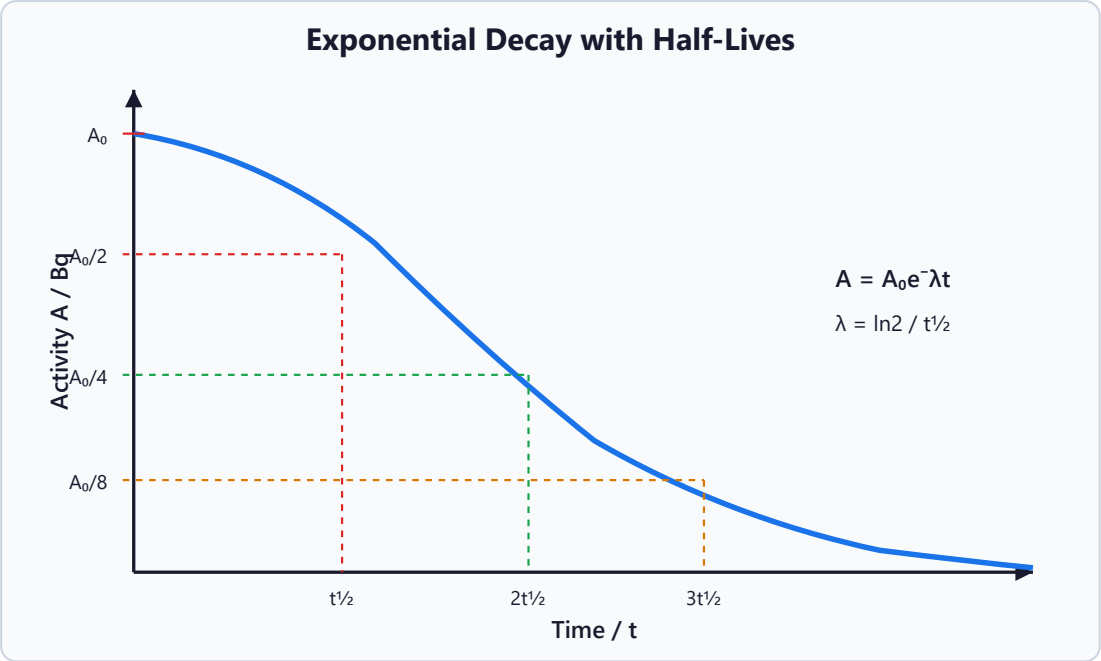
Activity (A): The number of decays per unit time (measured in becquerels, Bq, where 1 Bq = 1 decay per second).
Decay constant (λ): The probability of a nucleus decaying per unit time. Units: s^{-1}.
Half-life ($t_{½}$): The time taken for half the nuclei in a sample to decay, or for the activity to halve.

$$A = \lambda N$$

$$N = N_0 \, e^{-\lambda t}$$

$$A = A_0 \, e^{-\lambda t}$$

$$\lambda = \ln 2 \, / \, t_{½}$$



Exam Tip: To find the half-life from a decay graph, read the time taken for the activity (or count rate or number of nuclei) to halve from any starting point — not just the initial value. A graph of ln(A) vs t gives a straight line with gradient = −λ.

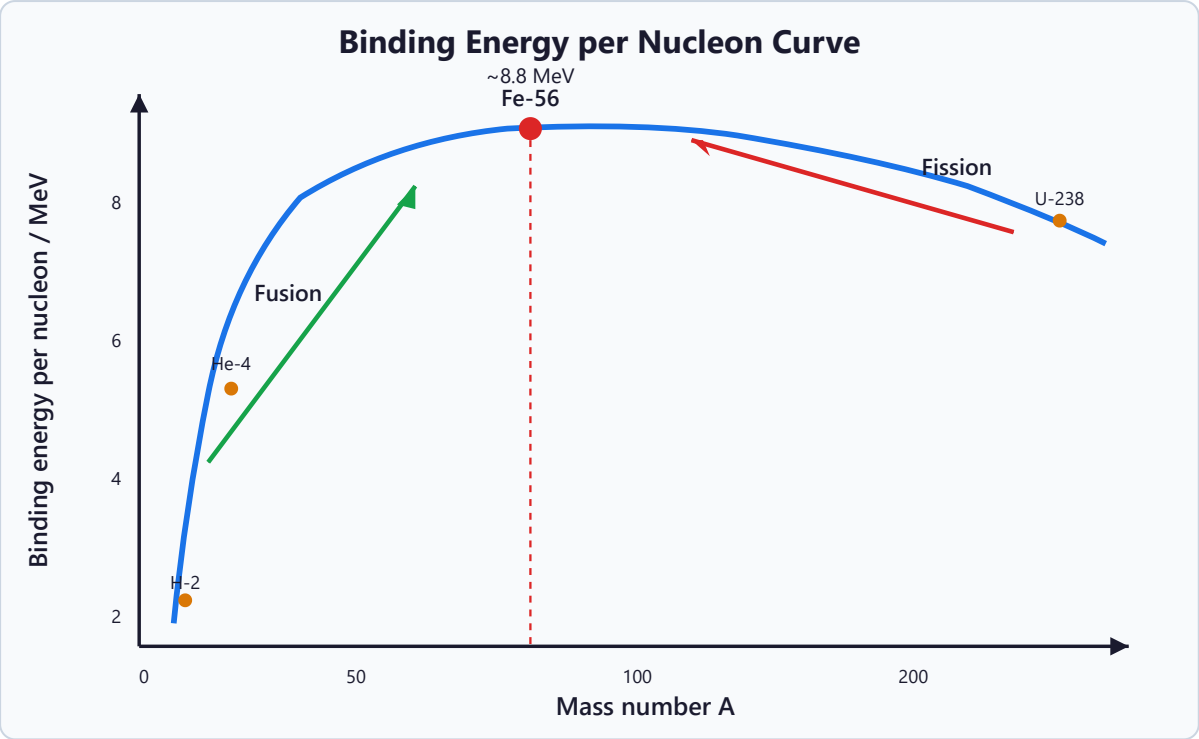
Binding Energy and Mass Defect

Mass defect (Δm): The difference between the mass of the separate nucleons and the mass of the assembled nucleus. The assembled nucleus has less mass.
Binding energy: The energy required to completely separate all the nucleons in a nucleus. Equivalently, the energy released when the nucleus is assembled from its nucleons.

$$E = \Delta mc^2$$

Where Δm = mass defect (kg), c = speed of light (3.00×10^8 m s^{-1}).

1 u = 1.661×10^{-27} kg = 931.5 MeV/c^2.

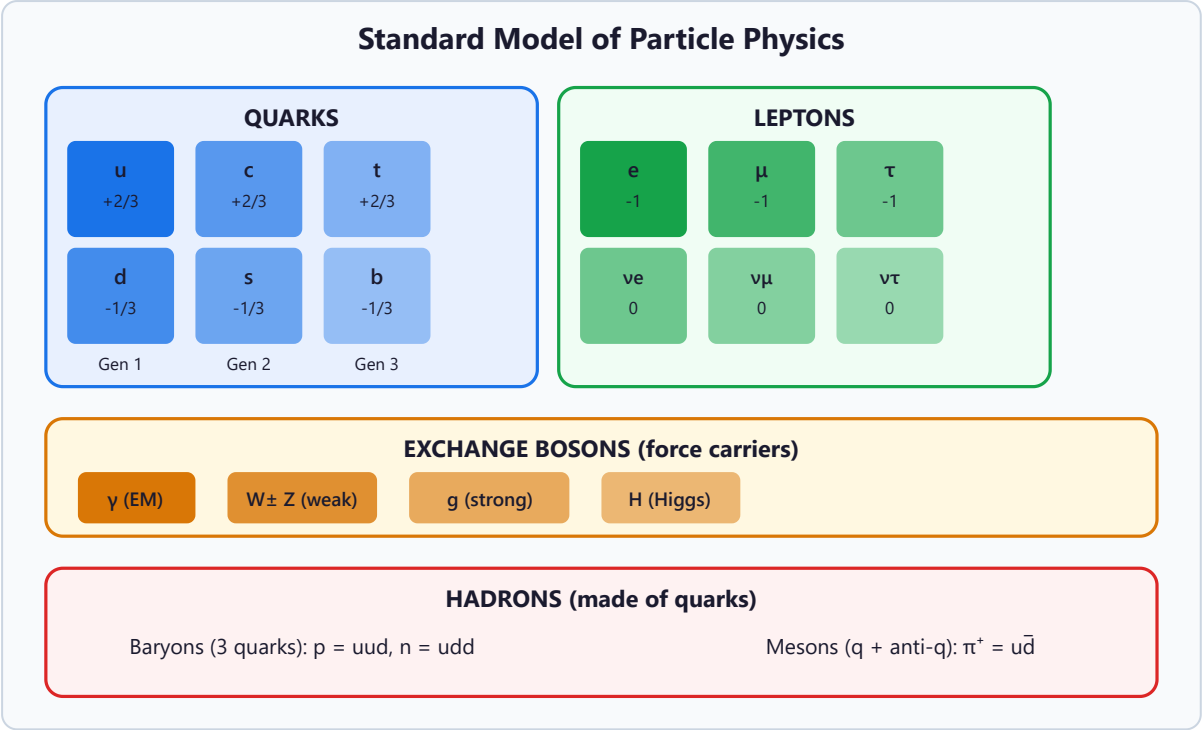


Binding energy per nucleon:

- A graph of binding energy per nucleon vs mass number shows a peak at iron-56 (~8.8 MeV per nucleon)
- **Fusion:** Light nuclei (below iron) combine to form heavier nuclei, releasing energy (binding energy per nucleon increases). Powers stars.
- **Fission:** Heavy nuclei (above iron) split into lighter nuclei, releasing energy (binding energy per nucleon increases). Used in nuclear power and weapons.

6.5 Particle Physics

The Standard Model



The Standard Model classifies all known fundamental particles:

Category	Particles	Properties
Quarks (6 flavours)	up (u), down (d), charm (c), strange (s), top (t), bottom (b)	Have fractional charges. Feel strong nuclear force. Always found in combinations (confinement).
Leptons (6 types)	electron (e^-), muon (μ^-), tau (τ^-), plus their neutrinos (ν_e, ν_μ, ν_τ)	Do not feel the strong force. Electron, muon, tau have charge −1e; neutrinos are neutral.
Exchange bosons	Photon (γ), $W^\pm$, Z^0, gluons, Higgs boson	Mediate fundamental forces.

Quark Properties

Quark	Charge	Baryon Number	Strangeness
up (u)	+2/3 e	+1/3	0
down (d)	−1/3 e	+1/3	0

strange (s)	−1/3 e	+1/3	−1
charm (c)	+2/3 e	+1/3	0
Antiquarks	All quantum numbers are reversed (e.g., anti-up has charge −2/3 e, baryon number −1/3)		

Hadrons

Hadrons: Particles made of quarks, that feel the strong nuclear force.
Baryons: Made of 3 quarks (e.g., proton = uud, neutron = udd). Baryon number = +1.
Antibaryons: Made of 3 antiquarks. Baryon number = −1.
Mesons: Made of a quark-antiquark pair (e.g., pion $\pi^+ = u\bar{d}$, kaon $K^+ = u\bar{s}$). Baryon number = 0.

Conservation Laws in Particle Interactions

Quantity	Always Conserved?
Energy and momentum	Always
Charge (Q)	Always
Baryon number (B)	Always
Lepton number (L_e, L_μ, L_τ)	Always (each flavour separately)
Strangeness (S)	Conserved in strong and EM interactions; can change by ±1 in weak interactions

Feynman Diagrams

- Feynman diagrams represent particle interactions. Time runs upwards (or left to right).
- Straight lines represent fermions (quarks, leptons). Wavy lines represent photons. Dashed or wavy lines represent W/Z bosons.
- At each vertex, charge, baryon number, and lepton number are conserved.

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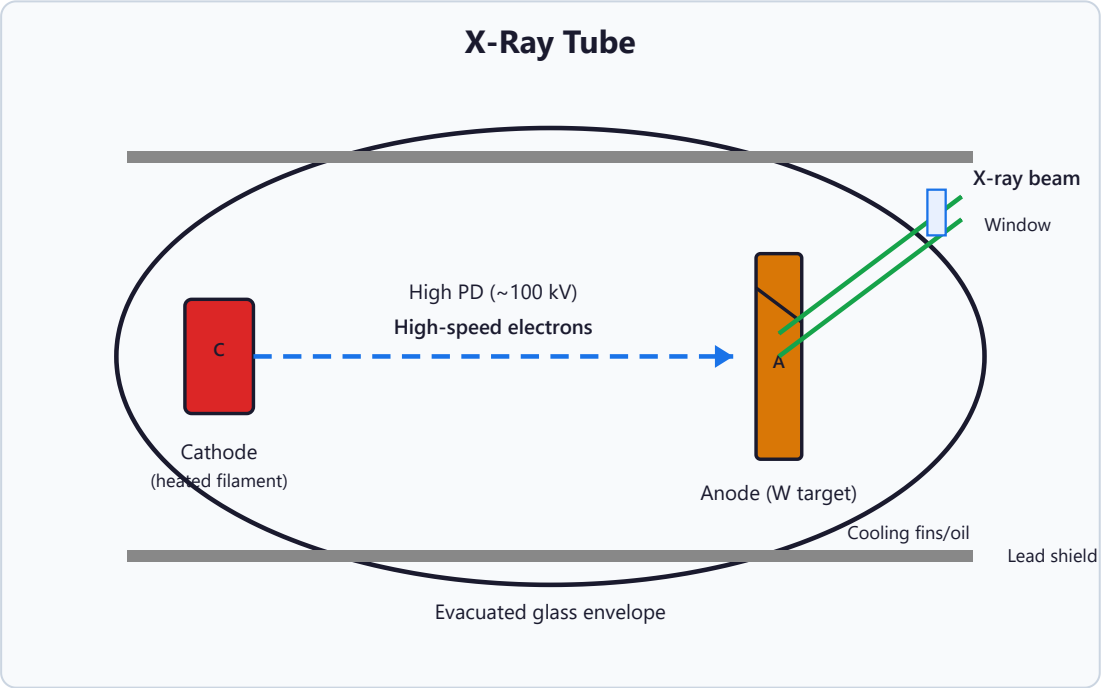
- The $W^{\pm}$ boson mediates the weak interaction (e.g., beta decay). The photon mediates the electromagnetic interaction. Gluons mediate the strong interaction.

Beta-minus decay Feynman diagram: A down quark emits a W^- boson and becomes an up quark. The W^- decays into an electron and an electron antineutrino.

> **Exam Tip:** When checking if a particle interaction is allowed, verify all conservation laws. If any one quantity is not conserved, the interaction cannot occur via that force. Remember strangeness can change in weak interactions but not strong ones.

6.6 Medical Imaging

X-rays



- Produced when high-speed electrons decelerate rapidly upon hitting a metal target (usually tungsten)
- X-ray images work because different tissues absorb X-rays differently: bone absorbs strongly (appears white), soft tissue absorbs moderately, air absorbs weakly (appears dark)
- Attenuation: $I = I_0 e^{-\mu x}$, where μ = linear attenuation coefficient (depends on material and X-ray energy), x = thickness

- Contrast media (e.g., barium, iodine) can be ingested or injected to highlight soft tissues
- Risk: ionising radiation — doses are kept as low as reasonably achievable (ALARA)

CT (Computed Tomography) Scans

- Multiple X-ray beams are taken from many angles around the body
- A computer reconstructs cross-sectional (slice) images, which can be combined into a 3D image
- Much higher radiation dose than a single X-ray
- Better at distinguishing soft tissues than conventional X-rays
- Used for: detecting tumours, internal injuries, brain haemorrhages, guiding biopsies

Ultrasound

> **Ultrasound:** Sound waves with frequencies above the range of human hearing (> 20 kHz). Medical ultrasound typically uses 1–15 MHz.

- A piezoelectric transducer converts electrical signals to ultrasound pulses and detects reflections
- Ultrasound is partially reflected at boundaries between tissues with different acoustic impedances
- **Acoustic impedance:** $Z = \rho c$, where ρ = density, c = speed of sound in the medium
- The fraction of intensity reflected at a boundary depends on the difference in acoustic impedances. Greater difference = more reflection.
- A coupling gel is used to prevent reflection at the skin-air boundary (air has very different Z from tissue)
- Distance to boundary: $d = ct/2$, where t = time for echo to return
- Advantages: non-ionising, real-time imaging, safe for foetus monitoring
- Disadvantages: lower resolution than X-ray/CT, poor at imaging bone and air-filled structures

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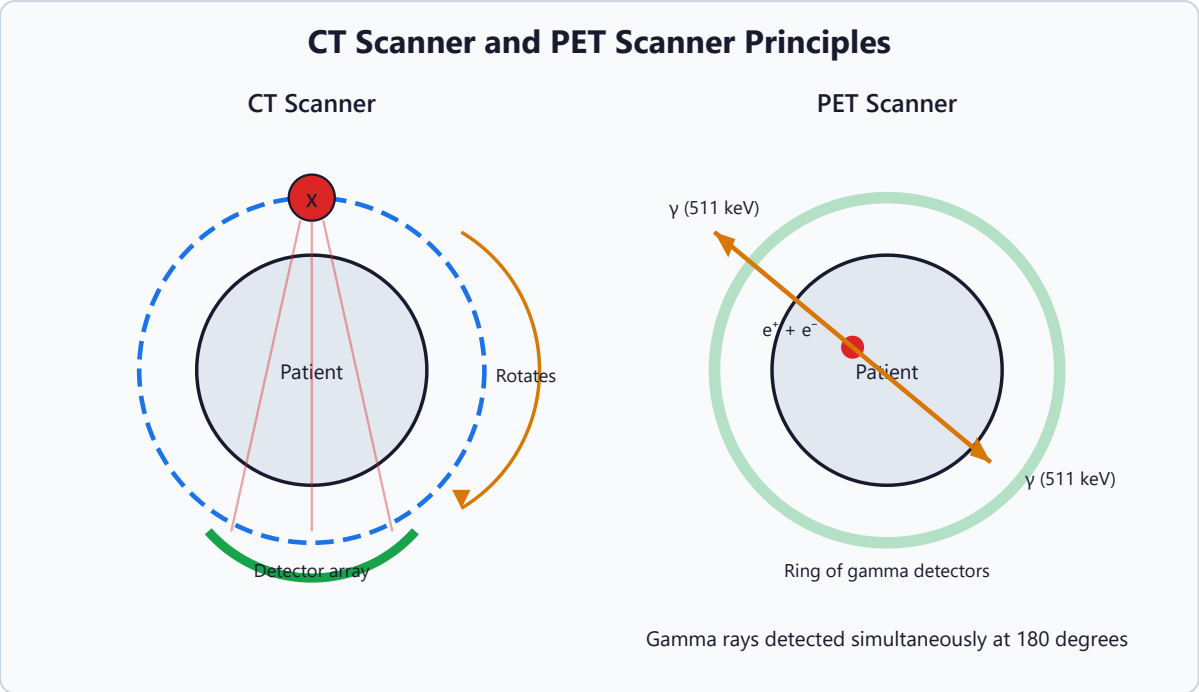

PET (Positron Emission Tomography) Scans

- A radioactive tracer that emits positrons (beta-plus emitter) is injected into the patient, typically fluorodeoxyglucose (FDG) labelled with fluorine-18
- The tracer accumulates in metabolically active tissues (e.g., tumours, brain regions)
- The positron travels a short distance then annihilates with an electron, producing two gamma photons travelling in opposite directions (180° apart), each with energy 511 keV (from $E = mc^2$)
- A ring of gamma detectors records coincident pairs of gamma photons. The line between the two detectors passes through the annihilation point.
- A computer reconstructs a 3D image of tracer concentration
- Used for: detecting cancer, monitoring brain activity, assessing heart function
- Often combined with CT (PET-CT) for both functional and structural imaging

Comparison of Medical Imaging Techniques

Feature	X-ray	CT	Ultrasound	PET
Radiation type	X-rays (ionising)	X-rays (ionising)	Sound waves (non-ionising)	Gamma rays (ionising)

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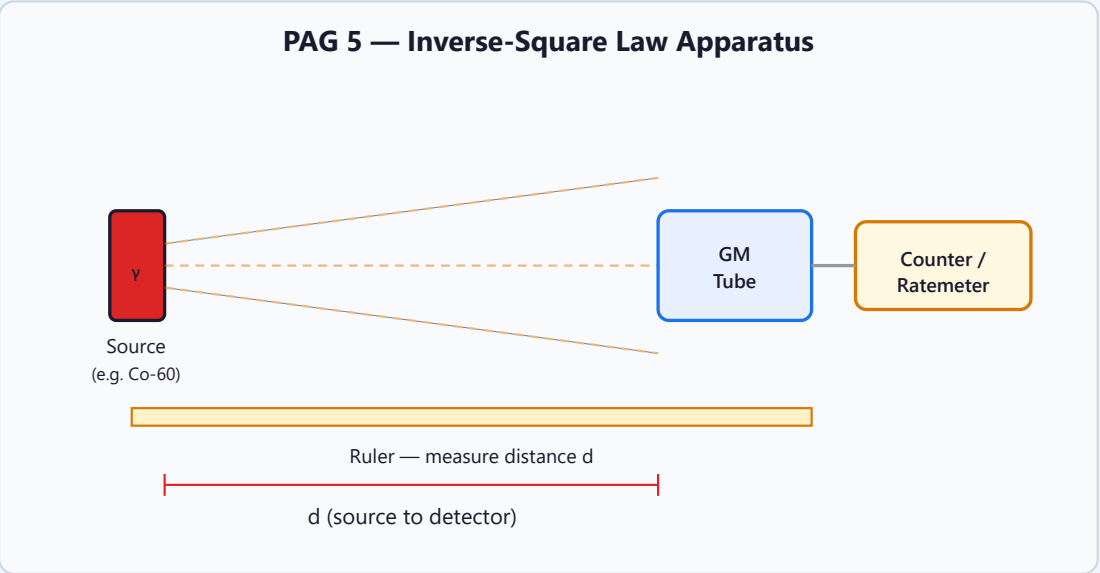
Image type	2D shadow	3D cross-sections	2D real-time	3D functional
Best for	Bones, chest	Soft tissue detail, brain	Foetus, soft tissue, blood flow	Cancer, brain activity
Resolution	Good	Very good	Moderate	Lower spatial; high functional
Radiation risk	Low dose	Higher dose	None	Moderate (tracer + gamma)

Exam Tip: In PET scans, the two gamma photons from electron-positron annihilation each have energy exactly 511 keV because the rest mass of an electron (and positron) is 0.511 MeV/c^2. The detection of simultaneous arrivals on opposite sides of the detector ring allows precise localisation.

⚗ PAG 5: Inverse-Square Law for Gamma Radiation



Apparatus

- Sealed gamma source (e.g. cobalt-60), handled with tongs
- Geiger-Muller (GM) tube connected to a counter or ratemeter

- Metre ruler
- Source holder and clamp stand
- Lead absorbers (for shielding and background reduction)

Method

1. First, measure the background count rate (with source removed) for at least 60 s; calculate the background count rate
2. Place the gamma source at a measured distance d from the GM tube window
3. Record the count over a fixed time (e.g. 60 s) and calculate the corrected count rate (subtract background)
4. Repeat for at least 6 different distances (e.g. 5 cm to 50 cm)
5. Plot corrected count rate (y-axis) against $1/d^2$ (x-axis)
6. A straight line through the origin confirms the inverse-square law

Safety

- Handle the source with long tongs — never with bare hands
- Store the source in a lead-lined container when not in use
- Keep maximum distance from the source (inverse-square law reduces dose)
- Minimise exposure time; point the source away from people
- Wear a dosimeter badge; follow school radiation safety procedures

Analysis

The intensity of gamma radiation follows the inverse-square law: $I = k/d^2$. If the corrected count rate C is proportional to intensity, then $C = k/d^2$. Plotting C against $1/d^2$ should give a straight line through the origin.

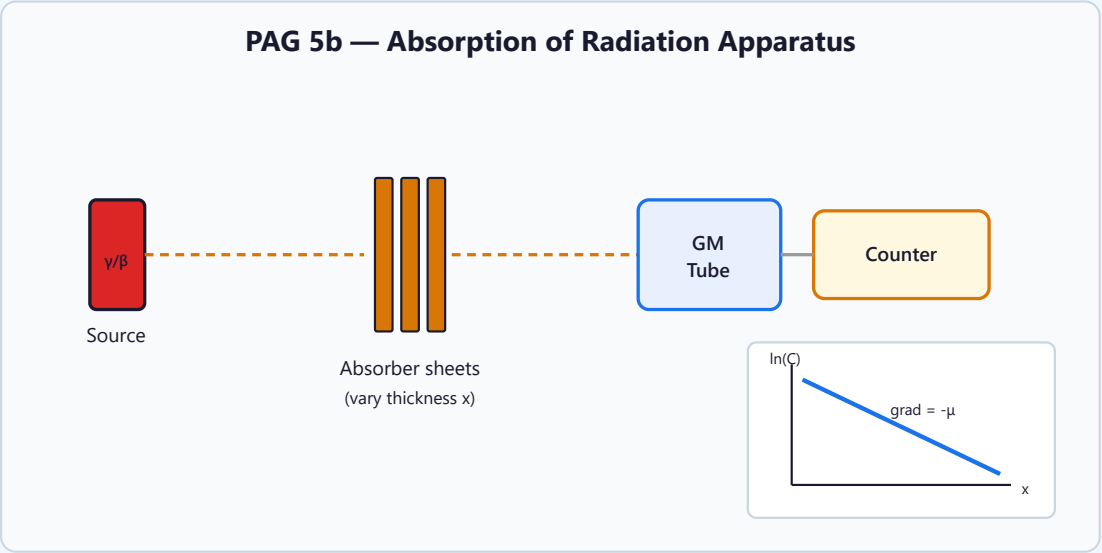
Sources of Error & Improvements

Source of Error	Type	Improvement

Radioactive decay is random — count rate fluctuates	Random	Count for longer time periods (e.g. 60 s minimum); repeat and average
Background radiation not accounted for	Systematic	Measure background count rate and subtract from all readings
Measuring d from wrong point on the GM tube	Systematic	Measure to the window of the GM tube; account for the dead space inside
Source may emit radiation in all directions (not a point source)	Systematic	Use large d compared to source size; align carefully

Exam Tip: Always subtract background count rate from your readings. If the graph of C vs $1/d^2$ does not pass through the origin, it may be because the effective centre of the GM tube is not where you measured d from (systematic zero error in d).

PAG 5b: Absorption of Radiation



Apparatus

- Gamma source (e.g. cobalt-60) or beta source, with tongs

- GM tube and counter/ratemeter
- Absorber sheets of varying thickness (e.g. lead sheets for gamma, aluminium for beta)
- Micrometer or vernier calipers to measure absorber thickness
- Ruler for distance measurement

Method

1. Measure background count rate (source removed, count for 60+ s)
2. Fix the source-to-detector distance (do not change this during the experiment)
3. Record the count rate with no absorber
4. Insert absorbers of increasing thickness x between the source and GM tube
5. For each thickness, record the count over a fixed time and calculate the corrected count rate C
6. Use at least 6 different thicknesses
7. Plot ln(C) against x — the gradient = $-\mu$ (linear attenuation coefficient)
8. The half-value thickness $x_{1/2} = \ln 2 / \mu$

Safety

- Handle radioactive source with long tongs only
- Store in a lead-lined container between uses
- Minimise exposure time; keep maximum distance
- Follow school/college radiation safety procedures

Analysis

Intensity decreases exponentially with thickness: $I = I_0 e^{-\mu x}$. Taking logs: $\ln(I) = \ln(I_0) - \mu x$. Since count rate C is proportional to I, plotting ln(C) vs x gives a straight line with gradient = $-\mu$ and y-intercept = $\ln(C_0)$.

Sources of Error & Improvements

Source of Error	Type	Improvement

Random nature of decay gives fluctuating count rates	Random	Count for longer periods (120+ s at low count rates); repeat measurements
Absorber thickness not uniform	Random	Use precision-machined absorbers; measure thickness at several points
Scattered radiation reaching the detector	Systematic	Use lead collimators to create a narrow beam
Background radiation not subtracted properly	Systematic	Measure background multiple times throughout the experiment

Exam Tip: The half-value thickness (HVT) is the thickness that reduces the intensity to half. After n half-value thicknesses, the intensity is $I_0/2^n$. This is analogous to half-life in radioactive decay.

Worked Example: A 220 μF capacitor is charged to 9.0 V and discharged through a 47 kΩ resistor. Calculate: (a) the time constant, (b) the voltage after 15 s.

(a) $\tau = RC = 47 \times 10^3 \times 220 \times 10^{-6} =$ **10.3 s**

(b) $V = V_0 e^{-t/RC} = 9.0 \times e^{-15/10.3} = 9.0 \times e^{-1.456} = 9.0 \times 0.233 =$ **2.1 V**

Worked Example: Verify the charge of the neutron from its quark composition (udd).

Charge = (+2/3) + (−1/3) + (−1/3) = +2/3 − 2/3 = **0** ✓
Baryon number = (+1/3) + (+1/3) + (+1/3) = **+1** ✓
Strangeness = 0 + 0 + 0 = **0** ✓